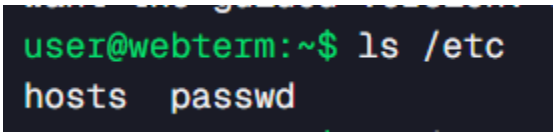
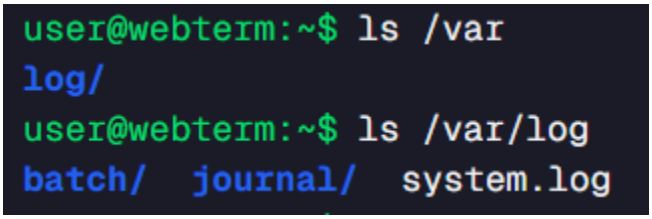
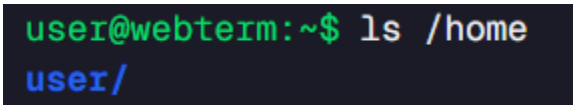
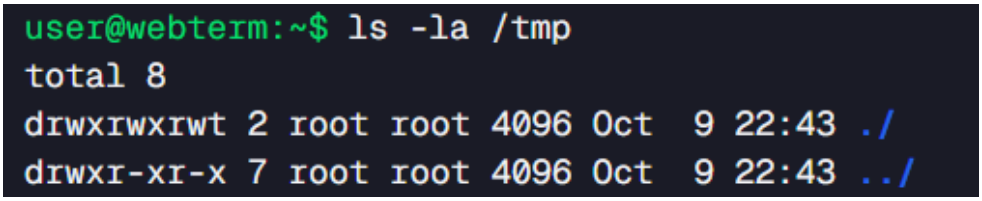
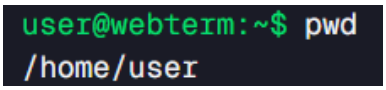
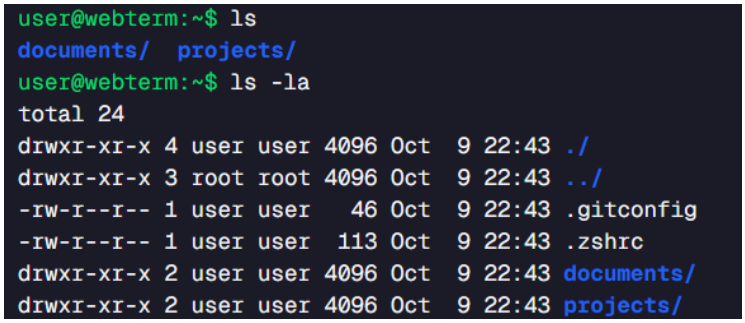
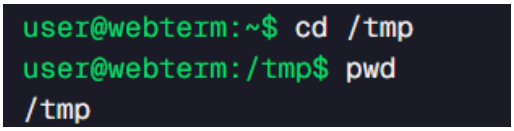
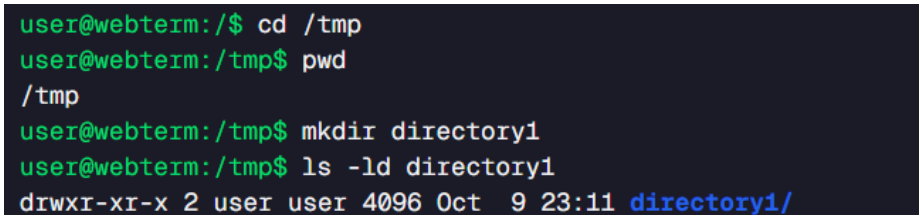
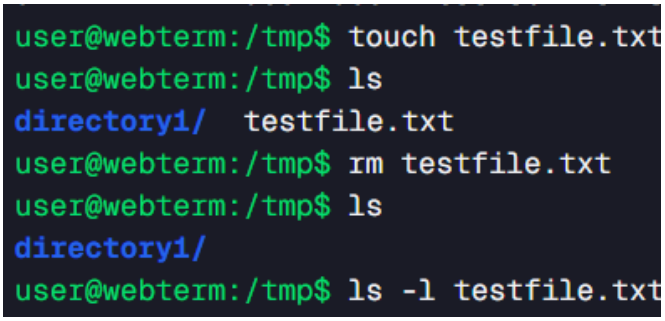
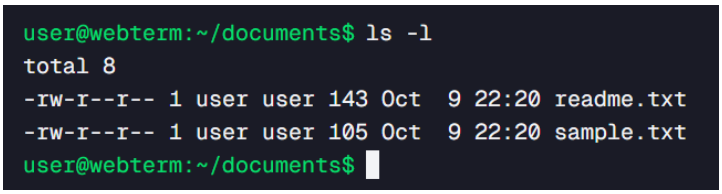
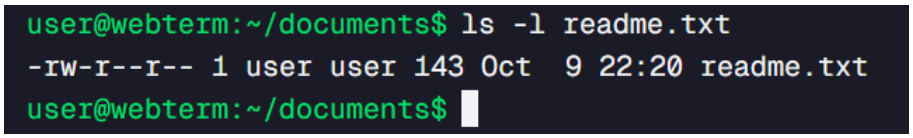
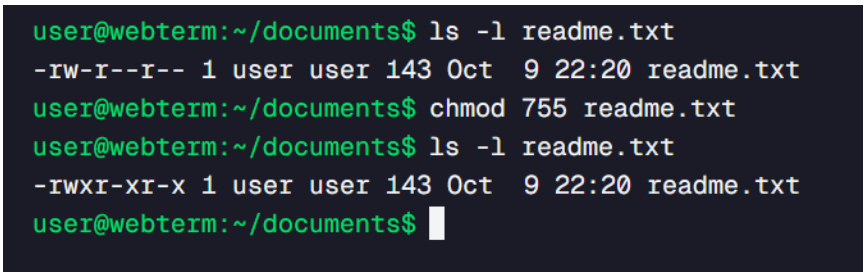
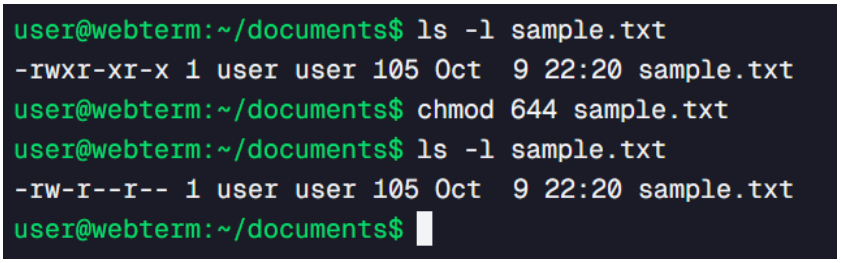
Linux File System Explorer

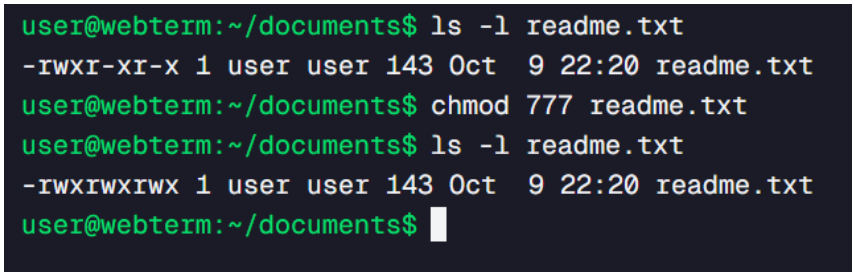
/etc	
/var	
/home	
/tmp	
/usr	

Linux Commands Investigation

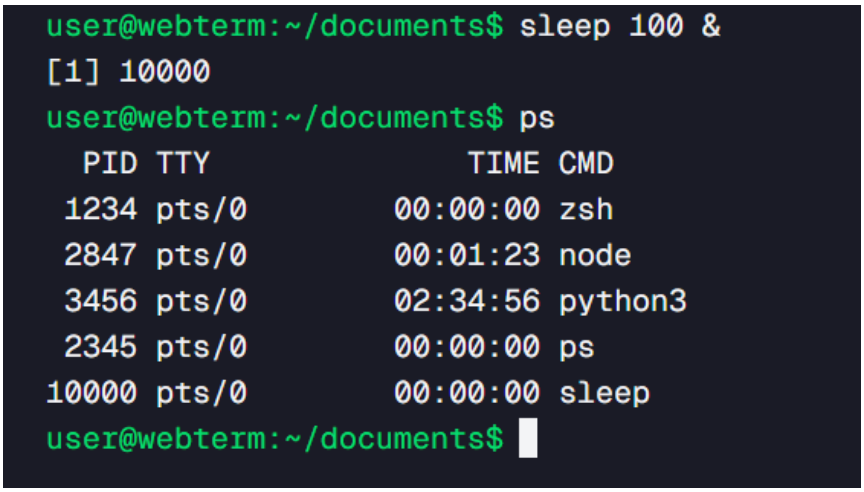
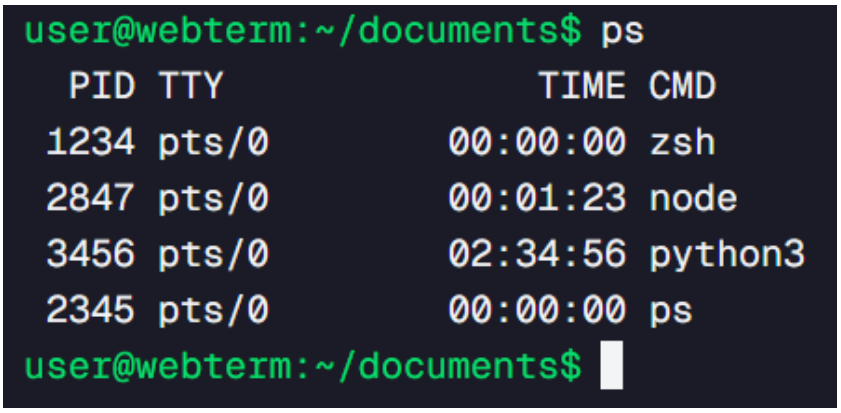
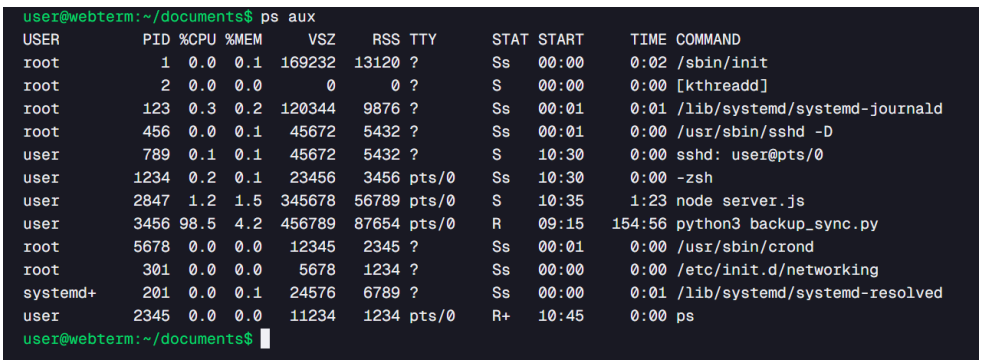
`pwd`	
`ls` and `ls -la`	
`cd`	
`mkdir`	
`rm`	

Linux Permissions

`ls -l`	
`ls -l readme.txt`	
`chmod 755`	
`Chmod 644`	

`chmod 777`	 user@webterm:~/documents$ ls -l readme.txt -rwxr-xr-x 1 user user 143 Oct 9 22:20 readme.txt user@webterm:~/documents$ chmod 777 readme.txt user@webterm:~/documents$ ls -l readme.txt -rwxrwxrwx 1 user user 143 Oct 9 22:20 readme.txt user@webterm:~/documents$

Linux Processes

Linux Processes	
`ps`	
`ps aux`	

`top -b -n 1`

```
user@webterm:~/documents$ top -b -n 1
top - 23:13:36 up 10:30:45, 2 users,  load average: 0.12, 0.15, 0.10
Tasks: 12 total,   2 running, 10 sleeping,   0 stopped,   0 zombie
%Cpu(s):  3.0 us,  1.3 sy,  0.0 ni, 95.4 id,  0.3 wa,  0.0 hi,  0.0 si,  0.0 st
MiB Mem :   7948.0 total,   1843.0 free,   3277.0 used,   2828.0 buff/cache
MiB Swap:   2048.0 total,   1920.0 free,   128.0 used.   4200.0 avail Mem

  PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM     TIME+ COMMAND
 3456 user      20   0  446.1m   85.6m   59.9m R  98.5   4.2   2:34:56 python3
 2345 user      20   0   11.0m    1.2m    987k R   1.5   0.0   0:00.01 top
 2847 user      20   0  337.6m   55.5m   38.8m S   1.2   1.5      1:23 node
  123 root      20   0  117.5m    9.6m    6.8m S   0.3   0.2      0:01 systemd-j
 1234 user      20   0   22.9m    3.4m    2.4m S   0.2   0.1      0:00 zsh
  789 user      20   0   44.6m    5.3m    3.7m S   0.1   0.1      0:00 sshd
    1 root      20   0  165.3m   12.8m    9.0m S   0.0   0.1      0:02 systemd
    2 root      20   0       0k      0k      0k S   0.0   0.0      0:00 kthreadd
  456 root      20   0   44.6m    5.3m    3.7m S   0.0   0.1      0:00 sshd
 5678 root      20   0   12.1m    2.3m    1.6m S   0.0   0.0      0:00 crond
  301 root      20   0    5.5m    1.2m    863k S   0.0   0.0      0:00 networking
  201 systemd-  20   0   24.0m    6.6m    4.6m S   0.0   0.1      0:01 systemd-r
user@webterm:~/documents$
```

Investigating Processes

```
user@webterm:~/documents$ ps aux
USER         PID %CPU %MEM     VSZ    RSS TTY      STAT START     TIME COMMAND
root           1  0.0  0.1  169232  13120 ?        Ss   00:00     0:02 /sbin/init
root           2  0.0  0.0       0      0 ?        S    00:00     0:00 [kthreadd]
root         123  0.3  0.2  120344   9876 ?        Ss   00:01     0:01 /lib/systemd/systemd-journald
root         456  0.0  0.1   45672   5432 ?        Ss   00:01     0:00 /usr/sbin/sshd -D
user         789  0.1  0.1   45672   5432 ?        S    10:30     0:00 sshd: user@pts/0
user        1234  0.2  0.1   23456   3456 pts/0    Ss   10:30     0:00 -zsh
user        2847  1.2  1.5  345678  56789 pts/0    S    10:35     1:23 node server.js
user        3456 98.5  4.2  456789  87654 pts/0    R    09:15   154:56 python3 backup_sync.py
root        5678  0.0  0.0   12345   2345 ?        Ss   00:01     0:00 /usr/sbin/crond
root         301  0.0  0.0    5678   1234 ?        Ss   00:00     0:00 /etc/init.d/networking
systemd+     201  0.0  0.1   24576   6789 ?        Ss   00:00     0:01 /lib/systemd/systemd-resolved
user        2345  0.0  0.0   11234   1234 pts/0    R+   10:45     0:00 ps
user       10000  0.0  0.1   12345   2345 pts/0    S    10:45     0:00 sleep 100
user       10001  0.0  0.1   12345   2345 pts/0    S    10:45     0:00 sleep 100
user       10002  0.0  0.1   12345   2345 pts/0    S    10:45     0:00 sleep 100
user       10003  0.0  0.1   12345   2345 pts/0    S    10:45     0:00 sleep 100
user@webterm:~/documents$ top -b -n 1
top - 23:19:58 up 10:30:45, 2 users,  load average: 0.12, 0.15, 0.10
Tasks: 12 total,   2 running, 10 sleeping,   0 stopped,   0 zombie
%Cpu(s):  3.0 us,  1.3 sy,  0.0 ni, 95.4 id,  0.3 wa,  0.0 hi,  0.0 si,  0.0 st
MiB Mem :   7948.0 total,   1843.0 free,   3277.0 used,   2828.0 buff/cache
MiB Swap:   2048.0 total,   1920.0 free,   128.0 used.   4200.0 avail Mem

  PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM     TIME+ COMMAND
 3456 user      20   0  446.1m   85.6m   59.9m R  98.5   4.2   2:34:56 python3
 2345 user      20   0   11.0m    1.2m    987k R   1.5   0.0   0:00.01 top
 2847 user      20   0  337.6m   55.5m   38.8m S   1.2   1.5      1:23 node
  123 root      20   0  117.5m    9.6m    6.8m S   0.3   0.2      0:01 systemd-j
 1234 user      20   0   22.9m    3.4m    2.4m S   0.2   0.1      0:00 zsh
  789 user      20   0   44.6m    5.3m    3.7m S   0.1   0.1      0:00 sshd
    1 root      20   0  165.3m   12.8m    9.0m S   0.0   0.1      0:02 systemd
    2 root      20   0       0k      0k      0k S   0.0   0.0      0:00 kthreadd
  456 root      20   0   44.6m    5.3m    3.7m S   0.0   0.1      0:00 sshd
 5678 root      20   0   12.1m    2.3m    1.6m S   0.0   0.0      0:00 crond
  301 root      20   0    5.5m    1.2m    863k S   0.0   0.0      0:00 networking
  201 systemd-  20   0   24.0m    6.6m    4.6m S   0.0   0.1      0:01 systemd-r
user@webterm:~/documents$
```